GEBREMEDHIN ASSEFA GIRMAY

AI & COMPUTATIONAL BIOLOGY | DEEP LEARNING | COMPUTER VISION | BIOINFORMATICS

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RESEARCH PROFILE

Computer Science researcher working at the intersection of artificial intelligence, deep learning, computer vision, bioinformatics, and computational biology. Research experience includes deep-learning methods for protein–ligand and protein–RNA interaction prediction, molecular representation learning, computational drug discovery, and semantic image segmentation. Interested in developing robust and generalizable AI methods for biomedical and scientific applications, with particular interest in PhD research in computer vision, biomedical image analysis, deep learning, and computational biology.

RESEARCH INTERESTS

Computer Vision • Deep Learning • Biomedical Image Segmentation • Medical Image Analysis • Image Classification • Object Detection • Semantic & Instance Segmentation • CNNs • ResNet • Vision Transformers • Transformer Architectures • Representation Learning • Transfer Learning • Protein–Ligand Interaction Prediction • Protein–RNA Interaction Prediction • Computational Drug Discovery • Molecular Representation Learning • Biomedical AI • Multimodal Learning

EDUCATION

Central South University, Changsha, China	M.Sc. Computer Science and Technology 2022–2026 GPA: 3.70 / 4.00
Master's Thesis: Protein-Ligand and Protein-RNA Interaction Prediction Based on Deep Learning Methods	Focus: Deep Learning • Bioinformatics • Computational Biology
University of Electronic Science and Technology of China (UESTC), Chengdu, China	B.Sc. Computer Science and Technology 2018–2022 GPA: 3.86 / 4.00
Bachelor's Thesis: Semantic Image Segmentation Based on Deep Learning Methods	Focus: Computer Vision • Deep Learning • Image Segmentation

RESEARCH EXPERIENCE

- Conducted research on deep-learning approaches for protein–ligand and protein–RNA interaction prediction.
- Designed and evaluated hybrid CNN/ResNet–Transformer architectures for molecular interaction modeling and binding-affinity prediction.
- Applied dataset preprocessing, representation learning, model training and validation, hyperparameter optimization, and comparative evaluation.
- Investigated semantic image segmentation using deep-learning methods, establishing a foundation for computer vision and biomedical image analysis.
- Worked with biological, molecular, and image datasets using Python-based scientific computing and machine-learning frameworks.
- Prepared research manuscripts, technical documentation, and academic presentations.

SELECTED PUBLICATIONS

Gebremedhin Assefa Girmay, Qahtan Adnan Aljanabi, and Lei Deng. DeepBindAffinity: A ResNet-Augmented Hybrid CNN-Transformer for Protein-Ligand Binding Affinity Prediction. 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025), pp. 126–129, 2025. DOI: 10.1109/BIBM66473.2025.11356467.

Qahtan Adnan Aljanabi, Zhijian Huang, Gebremedhin Assefa Girmay, Zhengkang Wang, Yuanpeng Zhang, and Lei Deng. DSSA: Dual-Stream Synthetic Accessibility Framework for Organic Compounds. 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025), pp. 633–638, 2025. DOI: 10.1109/BIBM66473.2025.11356617.

SELECTED RESEARCH PROJECTS

- DeepBindAffinity. Hybrid deep-learning framework combining residual convolutional feature extraction with Transformer-based representation learning for protein–ligand affinity prediction.
- Protein–RNA Interaction Prediction. Deep-learning approaches for biological representation learning and predictive modeling of protein–RNA interactions.
- Semantic Image Segmentation Based on Deep Learning Methods. Bachelor's thesis focused on pixel-level image understanding using deep neural networks.
- DSSA. AI-based assessment of organic-compound synthetic accessibility for computational drug-discovery applications.

PERSONAL PROJECTS COMPLETED

Multi-Omics Integration for Disease Subtype Classification (2022–2023)

- Built a multimodal learning framework integrating genomic, transcriptomic, and proteomic data for identifying disease subtypes. Applied feature extraction, dimensionality reduction, and classification models to uncover biologically meaningful patterns.
- Dataset: TCGA multi-omics dataset, including RNA-seq, DNA methylation, CNV, and protein expression data across multiple cancer types.

Cell Counting with Deep Learning (2020–2021)

- Built a convolutional neural network (CNN) in TensorFlow to automatically count cells in microscopy images from the BBBC005 dataset.

- Developed an automated alternative to manual cell counting, demonstrating the application of computer vision and deep learning to biological image analysis.

Chat App Development (2021–2022)

- Developed and deployed an Android application for real-time chatting.
- Worked across multiple programming and web-development frameworks, including Angular, Node.js, CSS, HTML, Python, Java, C++, MySQL, and PHP; contributed primarily as a backend developer.

COMPUTER VISION & DEEP LEARNING

Computer Vision • Image Classification • Semantic Segmentation • Instance Segmentation • Object Detection • Image Preprocessing • Data Augmentation • Feature Extraction • CNNs • ResNet • U-Net • Encoder–Decoder Architectures • Vision Transformers (ViT) • Attention Mechanisms • Transfer Learning • Representation Learning • Model Evaluation • PyTorch • OpenCV • NumPy • Pandas • Scikit-learn • Matplotlib • Jupyter

BIOINFORMATICS & COMPUTATIONAL BIOLOGY

Protein–Ligand Interaction Prediction • Protein–RNA Interaction Prediction • Binding Affinity Prediction • Multi-Omics Analysis • Molecular Representation Learning • Computational Drug Discovery • Synthetic Accessibility Prediction • Biological Sequence Analysis • Molecular Data Analysis • Deep Learning for Biological Data

PROGRAMMING & RESEARCH TOOLS

Python • C • C++ • Java • PyTorch • TensorFlow • OpenCV • Scikit-learn • NumPy • Pandas • Matplotlib • Jupyter Notebook • Git • SQL • Linux • HTML • CSS • JavaScript • React • Angular • Node.js • PHP • MySQL

HONORS & AWARDS

- Full Scholarship — University of Electronic Science and Technology of China (UESTC). Full scholarship support for bachelor's degree study.
- Chinese Government Scholarship — Central South University. Scholarship support for master's degree study.
- Academic Awards. Recognition received for academic performance and achievement.
- Sports Awards & Prizes. Awards and prizes received for participation and achievement in sporting activities.

RESEARCH METHODS

Deep Neural Network Development • Model Architecture Design • Dataset Preprocessing • Feature Engineering • Data Augmentation • Model Training & Validation • Hyperparameter Optimization • Performance Evaluation • Experimental Comparison • Scientific Literature Review • Manuscript Preparation • Academic Presentation

LANGUAGES

English — Professional Working Proficiency • Chinese — HSK 4 • Amharic — Native/Fluent • Tigrinya — Native/Fluent

ACADEMIC PRESENTATION

IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025) — Research presentation on DeepBindAffinity, a ResNet-augmented hybrid CNN–Transformer framework for protein–ligand binding affinity prediction.